- Any device with computer algebra system is prohibited during the exam.
- Each answer must be clearly labeled on the answer sheet.
- Problems 1 to 5 will be graded only according to final answers.
- You need to provide clear arguments and computations for Problem 6 and Problem 7.

Fill in the blanks. Each answer counts for 5 points.

1. $\lim_{x\to 0^+}\left(1-\frac{\sin x}{x}\right)^{\frac{1}{\ln x}} =$ __(1)__.

2. Let $f(x)=\int_1^{x-1} e^{-(t-1)^2}\,dt$.

   (a) Find the Taylor series for e^{-x^2} centered at $a=0$. __(2)__.

   (b) Find the Taylor series for $f(x)$ centered at $a=2$. __(3)__.

   (c) $\lim_{x\to 2}\frac{f(x)-\arctan(x-2)}{(x-2)^4\ln(2x-3)} =$ __(4)__.

3. Suppose that near $(x,y)=(1,-1)$ the equation

$$e^{x^2+y}-\ln\left(\frac{x}{y^2}\right)=1$$

   defines y as a twice differentiable function of x which is denoted by $y(x)$.

   (a) The linearization of $y(x)$ at $x=1$ is __(5)__.

   (b) $y''(1) =$ __(6)__.

4. $\int_0^2 \sqrt{\frac{4-x}{x}}\,dx =$ __(7)__.

5. Suppose that $f(x,y,z)$ is differentiable near $(x,y,z)=(0,1,2)$ and $f(0,1,2)=10$. Assume that curves

$$\mathbf{r}_1(t)=(3t,e^{2t},2\cos t),$$
$$\mathbf{r}_2(t)=(\ln t,t^2,\frac{2}{t}),$$

   lie on the level surface $f(x,y,z)=10$ and $f_y(0,1,2)=\frac{3}{2}$.

   (a) The tangent plane to $f(x,y,z)=10$ at $(x,y,z)=(0,1,2)$ is __(8)__.

   (b) Let

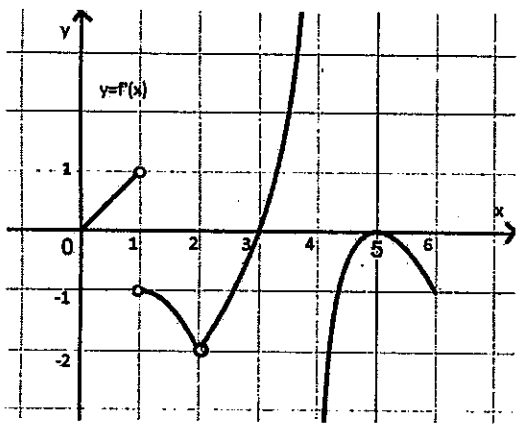
$$g(x,y,z)=\begin{cases}\dfrac{|z-2|\sin(\pi x)}{|x|+2|y-1|+|z-2|} & \text{, for } (x,y,z)\neq(0,1,2)\\ 0 & \text{, for } (x,y,z)=(0,1,2)\end{cases},$$

   and $\mathbf{u}=\frac{\nabla f(0,1,2)}{|\nabla f(0,1,2)|}$. Then $D_{\mathbf{u}}g(0,1,2) =$ __(9)__.

   (c) Assume that $f(x,y,z)$ attains the maximum value at $(x,y,z)=(0,1,2)$ when restricted to a level surface $h(x,y,z)=1$, where $h(x,y,z)$ is differentiable and $h_x(0,1,2)=6$. Using linear approximation, estimate the maximum value of $f(x,y,z)$ subject to the nearby level surface $h(x,y,z)=0.9$. The estimated value is approximately __(10)__.

見背面

6. Suppose that $f(x)$ is continuous on $[0,6]$ and $f(0) = f(4) = 0$. The graph of $f'(x)$ is given as below, but values of $f'(1)$, $f'(2)$ and $f'(4)$ are not determined. It is known that $\lim_{x\to 4^-} f'(x) = \infty$ and $\lim_{x\to 4^+} f'(x) = -\infty$.



(a) (5 pts) Find $\lim_{x\to 2^+} \dfrac{f(x)-f(2)}{x-2}$ and $\lim_{x\to 2^-} \dfrac{f(x)-f(2)}{x-2}$ by the Mean Value Theorem. Is $f(x)$ differentiable at $x = 2$? Justify your answers.

(b) (5 pts) List all critical numbers of $f(x)$ in the interval $(0,6)$. Where does $f(x)$ attain a local maximum? Where does $f(x)$ attain a local minimum?

(c) (8 pts) Find intervals on which $y = f(x)$ is concave upward. Find intervals on which $y = f(x)$ is concave downward. Find the inflection points of $f(x)$.

(d) (6 pts) Sketch the graph of $f(x)$ for $x \in [0,6]$.

(e) (6 pts) Suppose that $f'(x) = -\dfrac{32}{x^2+2x} - \dfrac{6}{x^2-6x+18}$, for $x > 6$. Find $\lim_{x\to\infty} f(x) - f(6)$.

7. Let S be the part of the graph

$$z = \cos x \sin y, \quad 0 \le x \le \frac{\pi}{2}, \ 0 \le y \le \pi,$$

with upward orientation. Consider the vector field $\mathbf{F}(x,y,z) = z\mathbf{i} + yz\mathbf{k}$.

(a) (10 pts) Parametrize the surface S and evaluate $\iint_S \mathrm{curl}\mathbf{F} \cdot d\mathbf{S}$ directly.

(b) (10 pts) Let C be the boundary curve of S. Decompose C as the union of differentiable parametric curves and evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ directly. Show that the line integral equals the surface integral in part (a) which is consistent with Stokes' Theorem.

試題隨卷繳回